

just an application. Remember IBM, Symantec and Verizon.

Put all this together and you have cloud computing, which is a very broad term covering a range of resources and services offered over the Internet. Some experts opine that any resource which a company uses over the Internet, outside its firewall, is 'on the cloud.' So, a company can choose, customise or develop an enterprise-wide solution, manage and maintain it, scale it up or down, or do whatever it wants, completely over the Internet or what we might say 'on the cloud,' which means the same thing. The cloud is nothing but a metaphor for the Internet, if that helps ease the confusion in any way!

The biggest advantage of cloud computing is that the end-user or developer does not need to bother about the physical location or configuration of the actual resources. It's for the service providers to worry about abstracting all the dirty details from the developers, who can work completely on a logical plane.

These systems can be easily scaled-up or down. You can start with a small server and a little storage during development, and then scale-up the volume or features of the system on-the-go, as the usage grows. Also, a company can use cloud resources simply to cater to temporary needs. They could even opt for a combination of their own resources along with the cloud's. In short, they can use what they want, and pay just for what they use.

Almost all literature on this subject compares the cloud to the electricity grid. You consume electricity without bothering about where it is generated and along which route it travelled to get to you. Nor do such details as how many others are using power from that grid bother you. Similarly, you can use software, services and infrastructure from the service providers using a Web interface, without worrying about any back-end details. Examples of cloud-computing platforms include Amazon's Elastic Compute Cloud (EC2), Salesforce's Force.com and Microsoft's Azure.

## Integration: An emerging service

So now, the cloud is raining resources. All we have to do is catch whatever we need in a tub, and start using it (and of course, pay for it too). But who does that? Cloud integration is no mean task, as it involves integrating not just many cloud-based resources but also integrating legacy non-cloud resources with the cloud-based ones.

Cloud-computing pioneers such as Amazon EC2 and Force.com have their own application platforms, in addition to the computing resources, so developers can easily build their applications on the platform and get going in no time. But that is just one of the simpler options. You can also do a complete custom build, by selecting disparate resources and putting these together, not to forget integrating the company's pre-cloud resources.

There are many integration tools available, including software-based, appliance-based, and cloud-delivered ones. Developers can choose amongst these integration tools to build their solutions. Cloud integration is emerging as a specialised genre of development and service. It is also a popular start-up idea.

The abstraction provided by cloud computing is so beneficial from a user's perspective, that some large organisations are cloaking even their existing infrastructure as 'private clouds.' What this means is that they consolidate all their resources and deliver it to various user-units within the company, just as cloud service providers would do to the public. The users within the company can use the resources they need without worrying about where they are, or how they are managed. To the company, it offers a two-fold benefit—user convenience, as well as better resource utilisation. No resource gets locked up. It is all on the private cloud, can be managed centrally, and provisioned as needed within the company. Plus, the life of resources also gets extended. An old, slow and steady computer can still be used, as long as it can connect to the network; it can run all the needed applications off the cloud.

"While the benefits of cloud computing are compelling, as with any product or technology, adoption takes its own course. Indian organisations, like many of their counterparts across the world, are in evaluation mode. They have largely moved away from the understanding mode, but adoption is not yet high. I think India presents an immense opportunity to realise the cloud possibilities. In developed countries, we see organisations looking at cloud adoption in stages—first, adopting infrastructure virtualisation;

then, setting up a private cloud; followed by selective use of the public cloud—before moving a large part of the application portfolio to the cloud (private or public).

For Indian organisations, there is a big opportunity to leapfrog to the cloud faster, given their underinvestment in information technology (IT). So they don't have to worry about managing legacy investments, creating an expensive migration plan, and then adopting the cloud," says Dhiraj Sinha, leader of the applications technology group, Dell Services. Many of Dell's solutions have used Open Source—for example, Dell is working with Canonical to help customers adopt Ubuntu-powered Open Source clouds, and the high-performance analytics services of Dell use Hadoop—the Open Source distributed computing and data-storage framework.

## "Open Source has built the cloud"

Sharing, the freedom to mix and match, choice, and many other characteristics of cloud computing demonstrate a likeness to the principles of Open Source software. More than just the similarities, there is also a sheer necessity for Open Source software and Open standards in clouds that comprise heterogeneous and often proprietary infrastructure. The availability of source code, freedom to modify and redistribute, flexibility and constant evolution, and other Open philoso-